

CS3213: Foundations of Software Engineering

Black-box Testing

Requirements/Documentation

substringsBetween

```
public static String[] substringsBetween(String str,  
                                         String open,  
                                         String close)
```

Searches a String for substrings delimited by a start and end tag, returning all matching substrings in an array.

A null input String returns null. A null open/close returns null (no match). An empty ("") open/close returns null (no match).

```
StringUtils.substringsBetween("[a][b][c]", "[", "]") = ["a", "b", "c"]  
StringUtils.substringsBetween(null, *, *)           = null  
StringUtils.substringsBetween(*, null, *)            = null  
StringUtils.substringsBetween(*, *, null)            = null  
StringUtils.substringsBetween("", "[", "]")          = []
```

Parameters:

str - the String containing the substrings, null returns null, empty returns empty

open - the String identifying the start of the substring, empty returns null

close - the String identifying the end of the substring, empty returns null

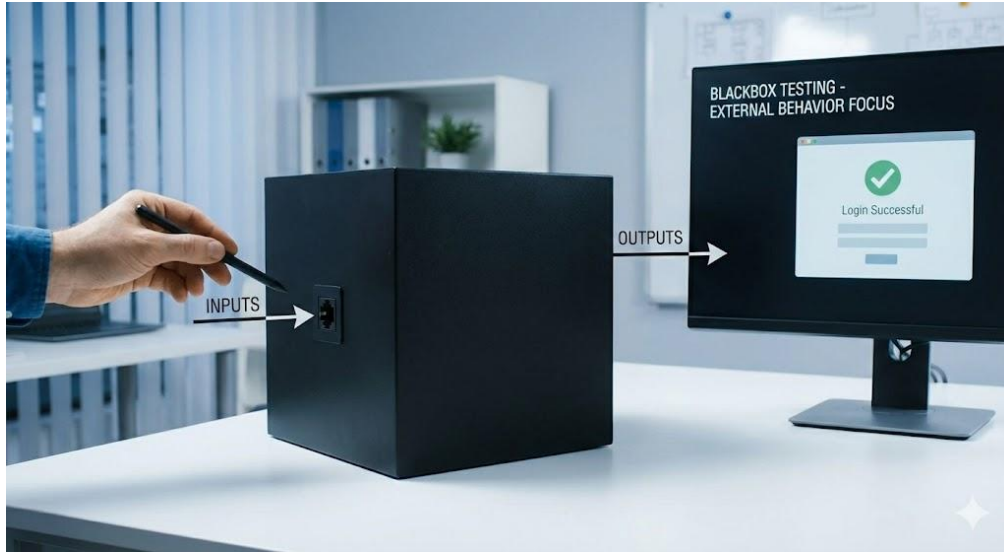
Returns:

a String Array of substrings, or null if no match

Since:

2.3

Black-box/Opaque-box Testing





Specification-based testing

- **Specification-based testing:** Derives tests based on the *requirements or documentation*
 - Agile user stories
 - Plan-driven use cases



Testing Workflow

1. Understand the requirements
2. Explore the program
3. Identify the partitions
4. Analyze the boundaries
5. Devise test cases
6. Automate test cases
7. Augment (creativity and experience)

1. Requirements/Documentation

substringsBetween

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```

Parameters:

str - the String containing the substrings, null returns null, empty returns empty

open - the opening tag

close - the closing tag

Returns:

a String

Since:

2.3

First: understand the requirements,
inputs, and outputs

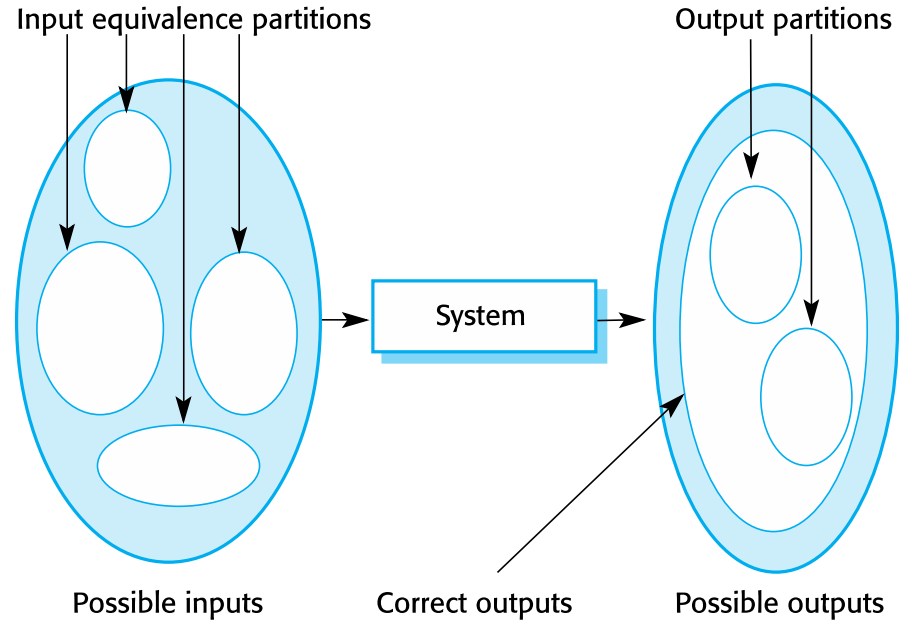
2. Exploratory Testing

```
@Test
void tests() {
    Assertions.assertArrayEquals(new String[] {"bc"},
        StringUtils.substringsBetween("abcd", "a", "d"));
    Assertions.assertArrayEquals(new String[] {"bc", "bc"},
        StringUtils.substringsBetween("abcdabcdab", "a", "d"));
}
```

Goal: explore what the program does to
increase your understanding and build a
mental model

3. Identifying Partitions

- Identify equivalent classes (partitions) of inputs (and outputs)
 - Consider each input individual
 - Consider each input with other inputs
 - Consider the different classes of output



3. Identifying Partitions: str

substringsBetween

```
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StringUtils.substringsBetween(*, *, null)            = null  
StringUtils.substringsBetween("", "[", "]")          = []
```

Parameters:

str - the String containing the substrings, null returns null, empty returns empty

open - the String identifying the start of the substring, empty returns null

close - the String identifying the end of the substring, empty returns null

Returns:

a String Array of substrings, or null if no match

Since:

2.3

Partitions for str:

- Null string
- Empty string
- String of length 1
- String of length > 1

3. Identifying Partitions: open

substringsBetween

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public static String[] substringsBetween(String str,  
                                       String open,  
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Searches a String for substrings delimited by a start and end tag, returning all matching substrings in an array.

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StringUtils.substringsBetween(*, *, null)            = null  
StringUtils.substringsBetween("", "[", "]")          = []
```

Parameters:

str - the String containing the substrings, null returns null, empty returns empty

open - the String identifying the start of the substring, empty returns null

close - the String identifying the end of the substring, empty returns null

Returns:

a String Array of substrings, or null if no match

Since:

2.3

Partitions for open:

- Null string
- Empty string
- String of length 1
- String of length > 1

3. Identifying Partitions: close

substringsBetween

```
public static String[] substringsBetween(String str,  
                                       String open,  
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Searches a String for substrings delimited by a start and end tag, returning all matching substrings in an array.

A null input String returns null. A null open/close returns null (no match). An empty ("") open/close returns null (no match).

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StringUtils.substringsBetween(*, *, null)            = null  
StringUtils.substringsBetween("", "[", "]")          = []
```

Parameters:

str - the String containing the substrings, null returns null, empty returns empty

open - the String identifying the start of the substring, empty returns null

close - the String identifying the end of the substring, empty returns null

Returns:

a String Array of substrings, or null if no match

Since:

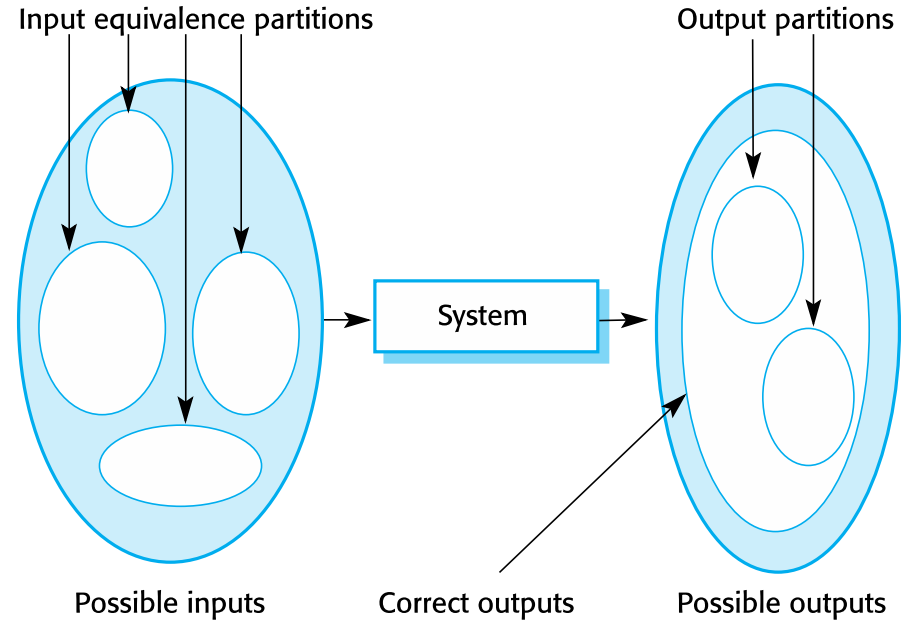
2.3

Partitions for close:

- Null string
- Empty string
- String of length 1
- String of length > 1

3. Identifying Partitions

- Identify equivalent classes (partitions) of inputs (and outputs)
 - Consider each input individual
 - Consider each input with other inputs
 - Consider the different classes of output



3. Identifying Partitions: Input Combinations

substringsBetween

```
public static String[] substringsBetween(String str,  
                                         String open,  
                                         String close)
```

Searches a String for substrings delimited by a start and end tag, returning all matching substrings in an array.

A null input

StringUtil
StringUtil
StringUtil
StringUtil
StringUtil

Parameters:

str - the S

open - the S

close - the

Returns:

a String Array of substrings, or null if no match

Since:

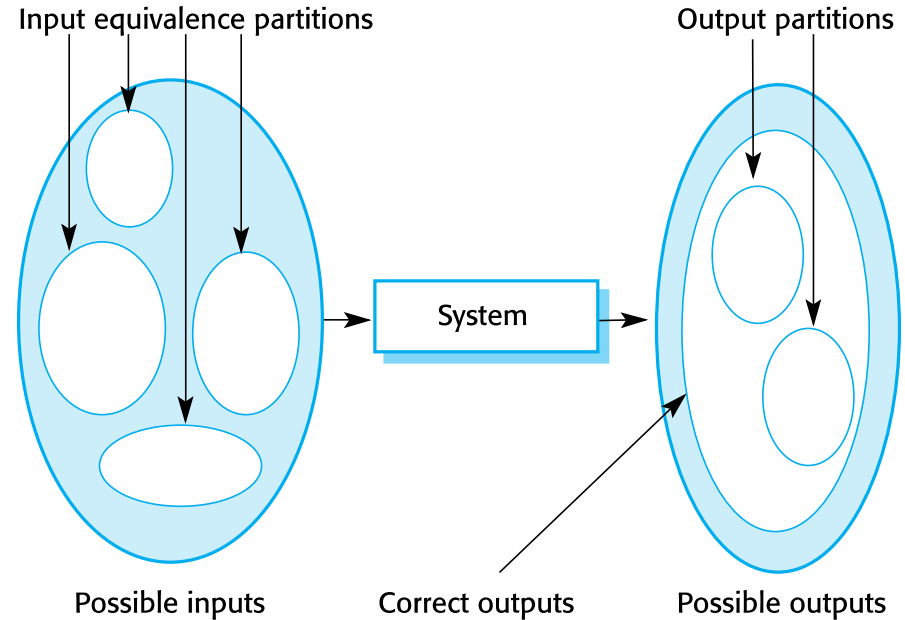
2.3

Combinations:

- Str contains neither the open nor the close tag
- Str contains the open tag, but not the close tag
- Str contains the close tag, but not the open tag
- Str contains both the open and close tags
- Str contains both the open and close tags multiple times

3. Identifying Partitions

- Identify equivalent classes (partitions) of inputs (and outputs)
 - Consider each input individual
 - Consider each input with other inputs
 - Consider the different classes of output



3. Identifying Partitions: Output Combinations

substringsBetween

```
public static String[] substringsBetween(String str,  
                                         String open,
```

Searcher

A null

String

String

String

String

String

Parameter

str -

open -

close

Returns:

a String

Since:

2.3

Combinations (Array of strings):

- Null array
- Empty array
- Single item
- Multiple item

Combinations (each individual string):

- Empty
- Single character
- Multiple characters

match).



3. Identifying Partitions: Output Combinations

- Input partitions are likely to exercise all the output partitions
- For complex programs, reflecting on the output partitions might help you identify input partitions that you did not identify before



Testing Workflow

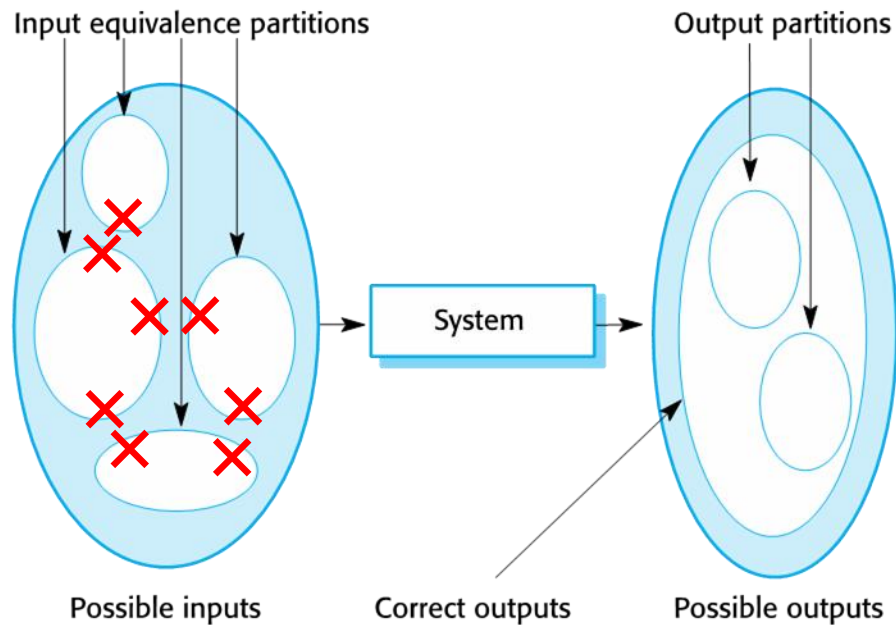
1. Understand the requirements
2. Explore the program
3. Identify the partitions
4. Analyze the boundaries
5. Devise test cases
6. Automate test cases
7. Augment (creativity and experience)



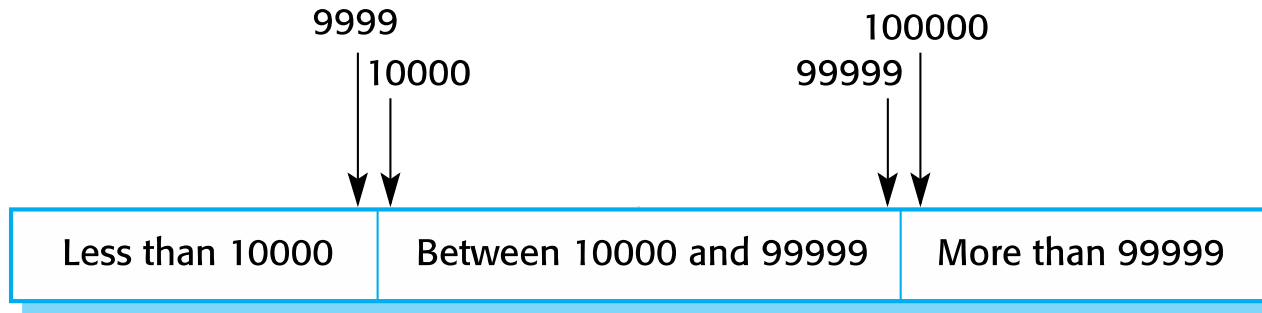
4. Analyzing Boundaries

- Inputs at *boundaries* are more likely to trigger bugs (e.g., off by one errors)
- Program behavior changes when going from one boundary to the other
- When identifying a boundary, write one test for each side of the boundary
 - Called *on point* and *off point* tests
 - On point: on the boundary
 - Off point: point closest to the boundary that belongs to the partition the on point does not belong to (i.e., the other partition)

4. Analyzing Boundaries



4. Analyzing Boundaries



Input values

4. Analyzing Boundaries

substringsBetween

```
public static String[] substringsBetween(String str,  
                                         String open,  
                                         String close)
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Searches a String for substrings delimited by a start and end tag, returning all matching substrings in a

A null input String returns null. A null open/close returns null (no match). An empty ("") open/clo

```
StringUtils.substringsBetween("[a][b][c]", "[", "]") = ["a","b","c"]
```

```
StringUtils.substringsBetween(null, *, *) = null
```

```
StringUtils.substringsBetween(*, null, *) = null
```

To cover the boundaries, we test
strings of length 2

Param

str - the String containing the substrings, null returns null, empty returns empty

open - the String identifying the start of the substring, empty returns null

close - the String identifying the end of the substring, empty returns null

Returns:

a String Array of substrings, or null if no match

Since:

2.3

Partitions for str:

- Null string
- Empty string
- String of length 1
- String of length > 1

5. Devising Test Cases

Identifying Partitions: str

```
substringsBetween
public static String[] substringsBetween(String str,
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                                       String close)
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A null input String returns null. A null open/close returns null (no match). An empty ("") open/close returns null (no match).

```
StringUtils.substringsBetween("[a][b]", "[", "]") = ["a","b",""]
StringUtils.substringsBetween(null, "", "") = null
StringUtils.substringsBetween("", "", "") = null
StringUtils.substringsBetween("", "", "[") = null
StringUtils.substringsBetween("", "[", "]") = []
```

Parameters:

str - the String containing the substrings, null returns null, empty returns empty
open - the String identifying the start of the substring, empty returns null
close - the String identifying the end of the substring, empty returns null

Returns:
a String Array of substrings, or null if no match

Since:
2.3

- Partitions for str:
- Null string
 - Empty string
 - String of length 1
 - String of length > 1

Identifying Partitions: open

```
substringsBetween
public static String[] substringsBetween(String str,
                                       String open,
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Parameters:

str - the String containing the substrings, null returns null, empty returns empty

open - the String identifying the start of the substring, empty returns null

close - the String identifying the end of the substring, empty returns null

Returns:

a String Array of substrings, or null if no match

Since:
2.3

Partitions for open:

- Null string
- Empty string
- String of length 1
- String of length > 1

4 x 4 x 4 x 5 = 320 potential tests

Identifying Partitions: close

```
substringsBetween
public static String[] substringsBetween(String str,
                                       String open,
                                       String close)
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```

Parameters:

str - the String containing the substrings, null returns null, empty returns empty

open - the String identifying the start of the substring, empty returns null

close - the String identifying the end of the substring, empty returns null

Returns:
a String Array of substrings, or null if no match

Since:
2.3

Partitions for close:

- Null string
- Empty string
- String of length 1
- String of length > 1

Identifying Partitions: Input Combinations

```
substringsBetween
public static String[] substringsBetween(String str,
                                       String open,
                                       String close)
```

Searches a String for substrings delimited by a start and end tag, returning all matching substrings in an array.

A null input

StringUtils

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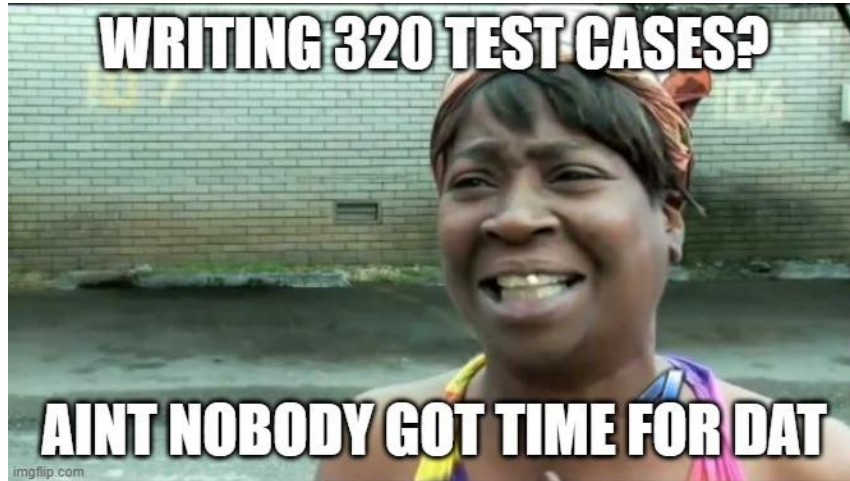
Combinations:

- Str contains neither the open nor the close tag
- Str contains the open tag, but not the close tag
- Str contains the close tag, but not the open tag
- Str contains both the open and close tags
- Str contains both the open and close tags multiple times

5. Devising Test Cases

- Partitions for str: Null string, Empty string, String of length 1, String of length > 1
- Partitions for open: Null string, Empty string, String of length 1, String of length > 1
- Partitions for close: Null string, Empty string, String of length 1, String of length > 1
- Combinations:
 - Str contains neither the open nor the close tag
 - Str contains the open tag, but not the close tag
 - Str contains the close tag, but not the open tag
 - Str contains both the open and close tags
 - Str contains both the open and close tags multiple times

5. Devising Test Cases





Devising Test Cases

- Pragmatically decide which ones to test

First idea: Test exceptional cases only once and not combine them (e.g., null and empty string)

Cases Only Tested Once

- Partitions for str: Null string, Empty string, String of length 1, String of length > 1
- Partitions for open: Null string, Empty string, String of length 1, String of length > 1
- Partitions for close: Null string, Empty string, String of length 1, String of length > 1
- Combinations:
 - Str contains neither the open nor the close tag
 - Str contains the open tag, but not the close tag
 - Str contains the close tag, but not the open tag
 - Str contains both the open and close tags
 - Str contains both the open and close tags multiple times



Cases Only Tested Once

- T1: str is null
- T2: str is empty
- T3: open is null
- T4: open is empty
- T5: close is null
- T6: close is empty

Cases Only Tested Once

- Partitions for str: String of length 1, String of length > 1
- Partitions for open: String of length 1, String of length > 1
- Partitions for close: String of length 1, String of length > 1
- Combinations:

- Str contains neither tag
- Str contains the open tag
- Str contains the close tag, but not the open tag
- Str contains both the open and close tags
- Str contains both the open and close tags multiple times

Second idea: For a str of length 1, there is no need to test for open length > 1 and close length > 1

$2 \times 2 \times 2 \times 5 = 40$ remaining potential tests

Remaining Cases

- Partitions for str: String of length 1, String of length > 1
- Partitions for open: String of length 1, String of length > 1
- Partitions for close: String of length 1, String of length > 1
- Combinations:
 - Str contains neither the open nor the close tag
 - Str contains the open tag, but not the close tag
 - Str contains the close tag, but not the open tag
 - Str contains both the open and close tags
 - Str contains both the open and close tags multiple times

This leaves as with four test cases for a string of length 1



Cases with str length = 1

- T7: Single character in str matches neither the open, nor the close tag
- T8: Single character in str matches the open tag, but not the close tag
- T9: Single character in str matches the close tag, but not the open tag
- T10: Single character in str matches both the open and close tags

Remaining Cases

- Partitions for str: String of length > 1
 - Partitions for open: String of length 1, String of length > 1
 - Partitions for close: String of length 1, String of length > 1
 - Combinations:
 - Str contains both open and close tags (assumption that tags are uniformly handled)
 - Str contains both the open and close tags multiple times
- Third idea: For the tags (open/close), test open length = 1 and close length = 1 as well as open length > 1 and close length > 1



Cases with $\text{str length} > 1$ open length = 1, close = 1

- T11: str contains neither the open, nor the close tag
- T12: str contains the open tag, but not the close tag
- T13: str contains the close tag, but not the open tag
- T14: str contains both the open and close tags
- T15: str contains both the open and close tags multiple times

Remaining Cases

- Partitions for str: String of length > 1
- Partitions for open: String of length > 1
- Partitions for close: String of length > 1
- Combinations:
 - Str contains neither the open nor the close tag
 - Str contains the open tag, but not the close tag
 - Str contains the close tag, but not the open tag
 - Str contains both the open and close tags
 - Str contains both the open and close tags multiple times



Cases with str length > 1 open length > 1, close > 1

- T16: str contains neither the open, nor the close tag
- T17: str contains the open tag, but not the close tag
- T18: str contains the close tag, but not the open tag
- T19: str contains both the open and close tags
- T20: str contains both the open and close tags multiple times



Additional boundary case(s)

- T21: str contains both the open and close tags with no characters between them

6. Automated Test Cases

- Instantiate the abstract tests and implement them as unit tests

```
@Test
void noSubstringBetweenOpenAndCloseTags() {
    Assertions.assertArrayEquals(new String[] {""},
        StringUtils.substringsBetween("aabb", "aa", "bb"));
}
```



7. Augmenting Test Suite

- Based on your experience, you might know
 - Whitespace characters can be an issue
 - Special characters
 - ...

Augment the test suite based on your
creativity and experience!

7. Augmenting Test Suite

```
@Test
void noSubstringBetweenOpenAndCloseTags() {
    Assertions.assertArrayEquals(new String[] {" b ", " "},
        StringUtils.substringsBetween("a b c a c", "a", "c"));
}
```



Recap: Testing Workflow

1. Understand the requirements
2. Explore the program
3. Identify the partitions
4. Analyze the boundaries
5. Devise test cases
6. Automate test cases
7. Augment (creativity and experience)

Sources and References

